Intermediate Elective 2

June McNally

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Set Up

Packages

```
library(tidyverse)
library(readxl)
library(ggtext)
```

```
library(tidyverse)
library(readxl)
library(ggtext)
library(here)
```

```
aquatic_data <- read_excel("../data/aquatic_data.xlsx")
taxon_list <- read_csv("../data/taxon_list.csv")
```

#Problems: 1. Explain Your Inspriation I chose a South China Morning Post–style timeseries line chart, inspired by Steven Ponce's "Coal falls. Renewables rise." visualization, which tracks how proportions of different categories change over time on a dark background with colored lines.

The Aquatic Insects sheet spans 2020–2025 with counts of invertebrate taxa across multiple sites, making a proportion timeseries ideal for showing how the relative community composition shifts year over year.

Year is mapped to the x-axis, proportion of total count to the y-axis, and taxon group (Copepod, Ostracod, Cladocera, Amphipod, Chironomid) to line color.

2. Plan Your Figure ## Sketch

![My figure sketIMG_6449.jpeg)

3.Code your figure

```
insects <- read_excel("../data/aquatic_data.xlsx",
                      sheet = "Aquatic Insects")
```

```
aquatic_data <- read_excel(here("data", "aquatic_data.xlsx"))
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
insects <- read_excel(here("data", "aquatic_data.xlsx"), sheet = "Aquatic Insects")
names(insects)
```

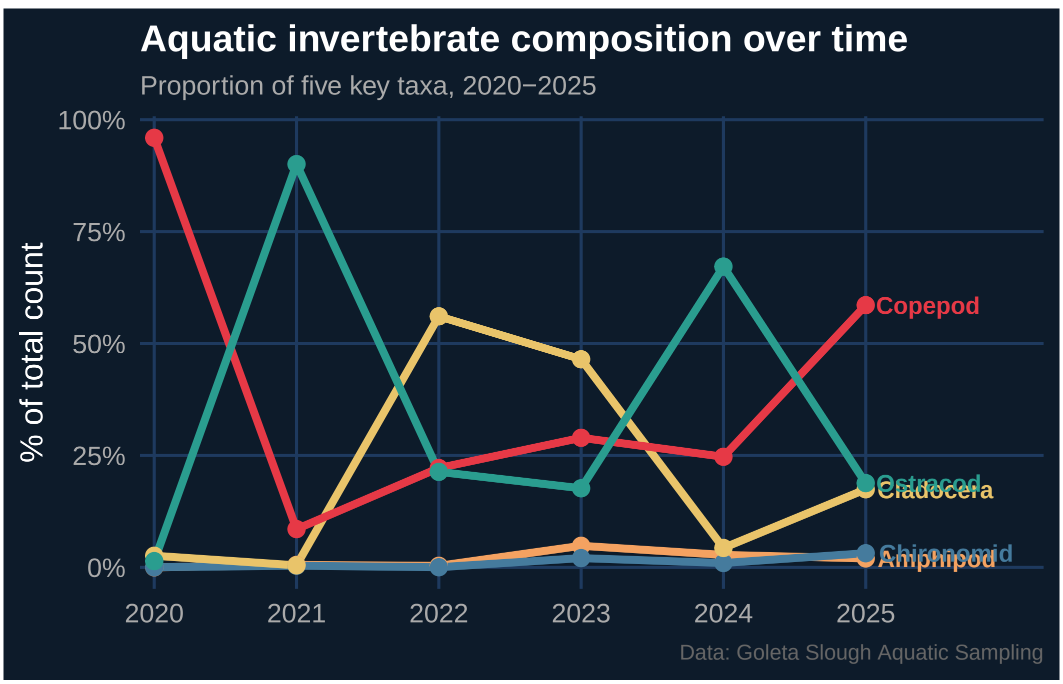
```
 [1] "Date data entered"
 [2] "Person Entering Data"
 [3] "Date on Vial"
 [4] "Person that sorted the sample"
 [5] "Site"
 [6] "Partial_sample_note"
 [7] "Sample Type"
 [8] "Ostracod"
 [9] "Copepod"
[10] "Hemiptera: Corixidae (boatman)"
[11] "Cladocera"
[12] "Nematode"
[13] "Diptera - TOTAL"
[14] "Dipteran sp."
[15] "Diptera: Chironomid"
[16] "Diptera: Ceratopogonidae"
[17] "Diptera : Ceratopogonidae pupa"
[18] "Diptera: Culicidae"
[19] "Diptera: Culicidae Larvae"
[20] "Diptera : Anthomyiidae"
[21] "Diptera: Ephydridae"
[22] "Diptera: Stratiomyidae"
[23] "Amphipod"
[24] "Lepidoptera-Crambidae"
[25] "Hemiptera  TOTAL"
[26] "Hemiptera- Mesoveliidae"
[27] "Hemiptera- Notonectidae"
[28] "Annelida TOTAL"
[29] "Annelida Sp."
```

```
[30] "Annelida: Oligochaete"
[31] "Annelida: Polychaete"
[32] "Ephemeroptera"
[33] "Odonate"
[34] "Coleoptera - TOTAL"
[35] "Coleoptera sp."
[36] "Coleoptera: Dytiscidae"
[37] "Coleoptera: Dytiscidae: Liodessus affinis"
[38] "Coleoptera: Dytiscidae: Agabus"
[39] "Coleoptera: Dytiscidae: Colymbetinae"
[40] "Coleoptera: Curculionidae  (Weavil)"
[41] "Coleoptera - Hydrophilidae"
[42] "Coleoptera - Tropiscernus"
[43] "Coleoperta Hydraenidae"
[44] "Coleoptera- Gyrindae- Gyrinini"
[45] "Ephemeroptera Sp."
[46] "Tipulidae Larva"
[47] "Plecoptera Sp."
[48] "Arachnida: Acari  (mite)"
[49] "Trichoptera Sp."
[50] "Collembola (Springtail)"
[51] "Gastropod - snail"
[52] "Mollusk"
[53] "Syrphidae"
[54] "Terrestrials"
[55] "Isopoda, sphaeroma"
[56] "Clam Shrimp"
[57] "Eristalis tenax larvae"
[58] "Larvae"
[59] "Crawfish"
[60] "Fish"
[61] "NZ Mudsnail"
[62] "Unknown"
[63] "Total #"
[64] "Comments"
```

```r
# Read the Aquatic Insects sheet from the Excel file
insects <- read_excel(here("data", "aquatic_data.xlsx"),
                      sheet = "Aquatic Insects")

# Define the five key taxa we want to visualize
key_taxa <- c("Copepod", "Ostracod", "Cladocera",
```

```r
              "Amphipod", "Diptera: Chironomid")

# Wrangle data: pivot to long format and calculate
# proportion of each taxon per year
plot_data <- insects %>%
  select(`Date on Vial`, all_of(key_taxa)) %>%
  rename(Date = `Date on Vial`) %>%
  filter(!is.na(Date)) %>%
  mutate(Date = as.Date(Date), Year = year(Date)) %>%
  pivot_longer(cols = all_of(key_taxa),
               names_to = "taxon", values_to = "count") %>%
  mutate(count = replace_na(count, 0),
         taxon = str_trim(taxon),
         taxon = if_else(taxon == "Diptera: Chironomid",
                         "Chironomid", taxon)) %>%
  group_by(Year, taxon) %>%
  summarise(total = sum(count, na.rm = TRUE), .groups = "drop") %>%
  group_by(Year) %>%
  mutate(proportion = total / sum(total) * 100) %>%
  ungroup() %>%
  filter(Year >= 2020, Year <= 2025)

# Define colors for each taxon
colors <- c("Copepod" = "#E63946", "Ostracod" = "#2A9D8F",
            "Cladocera" = "#E9C46A", "Amphipod" = "#F4A261",
            "Chironomid" = "#457B9D")

# Get the last year's values for direct line labeling
labels <- plot_data %>% filter(Year == max(Year))

# Build the plot
ggplot(plot_data, aes(x = Year, y = proportion, color = taxon)) +
  # Add lines and points for each taxon
  geom_line(linewidth = 1.4) +
  geom_point(size = 2.5) +
  # Add direct labels at end of each line instead of a legend
  geom_text(data = labels, aes(label = taxon),
            hjust = -0.1, size = 3.2, fontface = "bold") +
  # Apply custom colors
  scale_color_manual(values = colors) +
  # Format x and y axes
  scale_x_continuous(breaks = 2020:2025,
```

```r
                     expand = expansion(mult = c(0.02, 0.25))) +
  scale_y_continuous(labels = scales::label_number(suffix = "%")) +
  # Add titles and labels
  labs(title = "**Aquatic invertebrate composition over time**",
       subtitle = "Proportion of five key taxa, 2020-2025",
       x = NULL, y = "% of total count",
       caption = "Data: Goleta Slough Aquatic Sampling") +
  # Apply dark theme styling
  theme_minimal(base_size = 12) +
  theme(
    plot.background  = element_rect(fill = "#0D1B2A", color = NA),
    panel.background = element_rect(fill = "#0D1B2A", color = NA),
    text             = element_text(color = "white"),
    plot.title       = element_markdown(size = 14, color = "white"),
    plot.subtitle    = element_text(color = "#AAAAAA", size = 10),
    plot.caption     = element_text(color = "#666666", size = 8),
    axis.text        = element_text(color = "#AAAAAA"),
    panel.grid.major = element_line(color = "#1E3A5F"),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )
```



```r
# Save the final figure as a png file
ggsave(here("code", "aquatic_viz.png"), width = 8, height = 5, dpi = 300)
```

#Problems Continued 4. Describe your process

I adapted Steven Ponce's ggplot2 timeseries code by replacing his energy dataset with aquatic invertebrate sampling data from Goleta Slough, adjusting the color palette, and modifying the axis labels and annotations to fit the new context.

The biggest challenge was file path errors — getting R to find the data files correctly during rendering required switching from relative paths to the `here` package. I also had to find the exact column name for the Chironomid taxon since it had an unexpected trailing space.

Unlike simpler ggplot figures from class, this visualization uses a custom dark theme, direct line labeling instead of a legend, and `ggtext` for markdown formatting in the title — techniques I had not used before.